Q $y = \arcsin(x)$ $\frac{dy}{dx}$?

A $\arcsin(x) = y$

$\sin(y) = x$

$\frac{d}{dx}$ ↓

$\frac{d}{dx}(\sin(y)) = \frac{d}{dx}(x)$

$\cos(y) \cdot \frac{dy}{dx} = 1$

$\frac{dy}{dx} = \frac{1}{\cos(y)}$ ← Right side can't have y.

Side $\sin^2 y + \cos^2 y = 1$

$\cos^2 y = 1 - \sin^2 y$

$\cos y = \pm\sqrt{1-\sin^2 y}$

$\cos y = \sqrt{1-\sin^2 y}$

$\frac{dy}{dx} = \frac{1}{\sqrt{1-\sin^2 y}}$

$\frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}}$